PharmAchieve

PHARYNGITIS AND BRONCHITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care: Clinical Care	Providing Care: Distribution	Knowledge & Expertise
Communication & Collaboration	Leadership & Stewardship	Professionalism

LEGEND

Abx	Antibiotics
ADR	Adverse Drug Reaction
ARF	Acute Rheumatic Fever
COPD	Chronic Obstructive Pulmonary Disease
GAS	Group A Streptococcus
GABHS	Group A Beta-Hemolytic *Streptococcus*
GERD	Gastroesophageal Reflux Disease
GI	Gastrointestinal
OTC	Over-The-Counter
QTC	Heart-Rate Corrected QT Interval
RADT	Rapid Antigen Detection Test
RSV	Respiratory Syncytial Virus
SOB	Shortness of Breath
TB	Tuberculosis
RTI	Respiratory Tract Infection

DEFINITIONS

Upper RTI

Acute infections, typically viral but sometimes bacterial or rarely fungal, cause mild, self-limiting symptoms e.g. congestion, sore throat, and occasional fever

Acute rhinopharyngitis

Inflammation of the nasopharynx

Acute pharyngitis

Inflammation of the pharynx, commonly caused by a virus or, less commonly, group A *streptococcus*

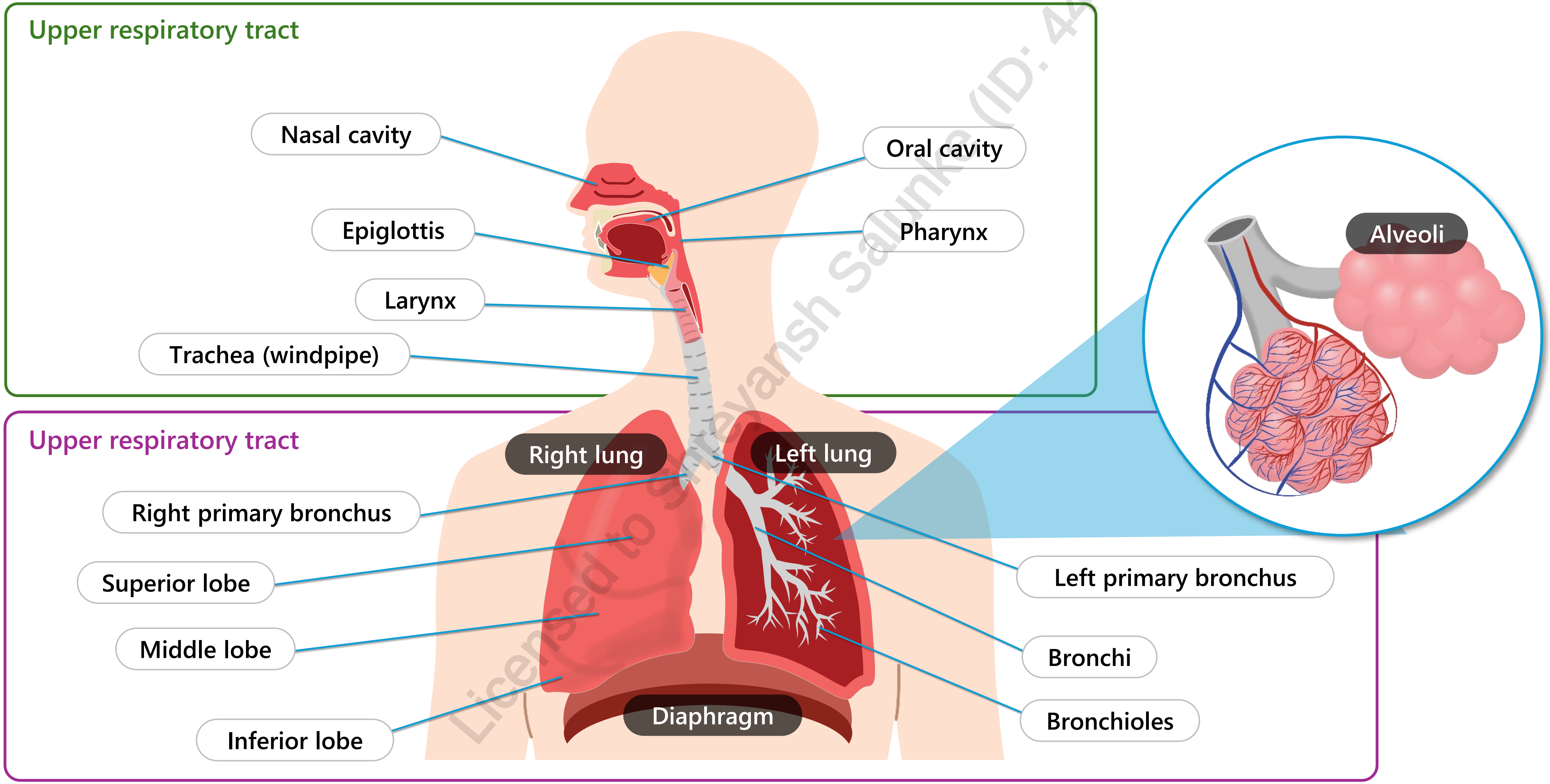
Lower RTI

Acute bronchitis

Inflammation of the bronchi, usually from viral infection but sometimes bacterial or due to irritant exposure (e.g. smoke)

ANATOMY OF UPPER RESPIRATORY TRACT

FYI



RISK FACTORS & PATHOGENESIS

- Exposure to others with infection
- Inadequate hand hygiene
- Crowded environments
- Exposure to lung irritants that compromise respiratory tract ciliary function: chemicals, smoking

Colonization of pathogens in URT → Aspiration to bronchae → Bronchial irritation and ciliary dysfunction → Clogged airways, mucus secretion and cough

DIFFERENTIAL DIAGNOSIS

FYI

- Allergies, asthma, cystic fibrosis, COPD (acute exacerbation)
- Irritants causing throat inflammation (e.g. smoking)
- Sinusitis, pneumonia, pertussis (whooping cough), tuberculosis
- Primary influenza infection and secondary influenza complication
- GERD

- Drug-related symptoms
  - Cough (ACEI)
  - SOB (beta-blockers)
  - Flu-like symptoms: vaccines, monoclonal antibodies, missed doses of antidepressants (withdrawal effects)

CLINICAL PRESENTATION

Acute pharyngitis

- Most cases of acute pharyngitis are viral and self-limiting Symptoms can include:
  - Low-grade fever, coughing, rhinorrhea, rash, headache, conjunctivitis
- GAS (strep throat), symptoms can include:
  - Sudden onset of sore throat, fever, absence of cough, palatal petechiae, and possible scarlet fever (rare)
  - Rare complications include invasive GAS, neck space infection, and glomerulonephritis

Acute bronchitis (chest cold)

Usually follows a viral URTI, adult symptoms are often self-limiting (10 – 14 days) and can include:

Coughing (+/- sputum production), URTI symptoms (e.g. nasal congestion, runny nose, watery eyes)

Cough can last 3 – 4 weeks (bronchitis often suspected following > 5 days of coughing)

Purulent sputum is not an indication of bacterial infection, but rather the presence of leukocytes and inflammation

Symptoms in children can include:

Wheezing, chills, chest congestion, vomiting, tachypnea, respiratory distress, hypoxemia (rare)

CLINICAL PRESENTATION & DIAGNOSIS

Usually self-limiting and resolves in 10-14 days (in some cases up to 3-4 weeks)

	Non-specific URTI	Bacterial Pharyngitis	Acute bronchitis
Clinical Symptoms	• Early: rhinitis, coryza, sore or scratchy throat, malaise, conjunctivitis • Delayed: nasal congestion, mild cough, nasal discharge • Uncommon: fever (sometimes present in children), nausea, vomiting	• Early: sudden onset severe sore throat, fever, headache, swollen cervical lymph nodes, Tonsillar exudate, nausea and vomiting • Delayed: scarlet skin rash, glomerulonephritis • Uncommon: Cough, coryza, nasal congestion	• Early: rhinitis, coryza, sore or scratchy throat, malaise, conjunctivitis • Delayed: persistent cough sputum production, wheezing • Uncommon: fever, nausea, vomiting • Early symptoms of acute bronchitis and URTI are the same
Diagnosis	• Presence of signs and symptoms without any severe or systemic symptoms	• Rapid antigen test (RAT) or throat culture is best way to diagnose GAS. • RAT ok to rule out adults, need culture for children	• Eliminate suspicion of pneumonia and influenza. • Presence of high fever and systemic signs such as increased breathing rate and tachycardia should prompt referral

ETIOLOGY

Acute Pharyngitis

Non-bacterial organisms

- Viral (rhinovirus, coronavirus, adenovirus) common

Bacterial Organisms

- Bacterial pathogens include *Streptococcus pyogenes* (Group A β-Hemolytic streptococci) which is also known as 'strep throat' and rarely others like *Fusobacterium*, *Neisseria*

Acute Bronchitis

Non-bacterial organisms

- Viral causes account for > 90% of cases - pathogens include rhinovirus, influenza A and B, parainfluenza, respiratory syncytial virus (RSV), coronavirus 1 to 3, enterovirus, and adenovirus

Bacterial Organisms

- Uncommon causes of bronchitis cases
- Bacterial sources include *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*, *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, *Bordetella pertussis*, *Bordetella parapertussis*

NON-PHARMACOLOGICAL MEASURES

Education

- Likely viral cause of URTI/bronchitis/pharyngitis
- NO benefit from antibiotics except in the cases of pertussis or acute pharyngitis secondary to bacterial pathogen. Risk of antibiotic resistance if antibiotics used
- Limiting spread of infection through proper hand washing
- Avoiding environmental irritants such as toxin/allergen exposure (tobacco smoke, pollen)

Over the counter

- Lozenges and lidocaine can be used to provide mild pain relief

Humidity

- Increase humidity to reduce cough

Comfort

- Improve patient's comfort by rest, activity as tolerated

Fluids

- Use fluids to prevent dehydration in children and reduce viscosity of respiratory secretions

NATURAL REMEDIES

FYI

- Echinacea: no conclusive evidence showing that it reduces duration or severity
- Honey: no evidence that it reduces the duration but may help with symptomatic treatment of cough or sore throat
  - Use pasteurized honey for symptom management
  - Use in children >1 years old.
- Vitamin C: no conclusive evidence that it reduces the duration or severity of URTI
- Nasal saline spray helps with symptomatic treatment of nasal congestion and discomfort
- Zinc: often available as lozenges. There is some evidence that zinc shortens the duration and severity of a cold. Intranasal zinc has the side effect of permanent anosmia. Lozenges can also cause mouth irritation- avoid in aphthous ulcers

PHARMACOLOGICAL MEASURES- SYMPTOMATIC RELIEF

- Analgesics and antipyretics
  - Acetaminophen and ibuprofen for fever, headache and pain
  - Avoid aspirin in children and adolescents due to risk of Reye's syndrome
- Antitussives
  - Dextromethorphan
  - Antitussives are not recommended for productive coughs
  - Avoid in children < 6 years of age: lack of evidence for efficacy and risk of adverse effects
  - As per Health Canada: recommended to NOT use opioid-containing products in <18 years old
  - Both have risk for abuse
- Bronchodilators, inhaled corticosteroids
  - Used in patients with wheezing; no evidence of efficacy unless airway obstruction is present
- Antihistamines
  - Avoid antihistamines as they cause drying of mucus secretions, which prevents effective clearance
- Mucolytics (guaifenesin)
  - No evidence of efficacy

BACTERIAL PHARYNGITIS

The Modified Centor Score estimates the probability of *streptococcal* pharyngitis based on symptoms and physical findings

- Guides clinicians on when to perform diagnostic testing and when empirical antibiotic treatment may be appropriate. It also helps to identify patients unlikely to benefit from antibiotics

Parameter	Points
Temperature > $38^{0}C$	1
Absence of cough	1
Swollen tender cervical nodes	1
Tonsillar swelling / exudate	1
Age	
3* to 14 years	1
15 to 44 years	0
≥45 years	-1

*GAS pharyngitis is rare in children < 3 years. Testing is only required during an outbreak or when scarlet fever is suspected

Score	Risk of GABHS	Suggested Management
≤ 2	1 – 17%	No further testing or antibiotics
≥ 3	28 – 53%	Perform RADT** • If positive → Treat with Abx • If negative & < 20 years → throat culture, if positive for GAS, then start Abx • If negative & ≥ 20 years → No antibiotics

**RADT is not sensitive in children (risk of false negatives). If RADT is negative in a child, then perform culture to confirm results
→ A throat culture is the gold standard

PHARMACOLOGICAL MEASURES - OVERVIEW

Bronchitis	• Routine antibiotic treatment is not recommended for acute bronchitis, since virus account for up to 95% cases • Consider further investigation: symptoms last >14 days without improvement or worsening, if linked to a pertussis case OR if immunocompromised
Bacterial pharyngitis	• Antibiotic therapy aimed at eradicating GAS is recommended for cases of bacterial pharyngitis confirmed by throat culture or antigen testing • Concurrent use of corticosteroids with antibiotics is not recommended

BACTERIAL PHARYNGITIS TREATMENT IN CHILDREN

- ❑ Treatment is indicated within 9 days of symptom onset
  - ❑ Optimal treatment against GAS is amoxicillin or penicillin for 10 days

Drugs	Indication	Notes
Penicillin V x 10 days Amoxicillin 50mg/kg per day x 10 days	Most children	• Excellent safety profile • May cause mild diarrhea
Penicillin G Benzathine x 1 dose	For children unlikely to complete 10-day antibiotic course	• Intramuscular injection
Amoxicillin challenge Cephalexin 20mg/kg/dose twice daily x 10 days	Non-anaphylactic hypersensitivity reactions to penicillin	• A low-risk patient is given a small initial dose of amoxicillin and observed for 20-30 minutes
Clarithromycin x 10 days Azithromycin x 5 days Clindamycin x 10 days	Documented type 1 hypersensitivity to penicillin (anaphylaxis)	• Verify local antimicrobial susceptibility due to high regional resistance rates

BACTERIAL PHARYNGITIS IN ADULTS

Drugs	Indication	Notes
Amoxicillin x 6-10 days Penicillin V x 10 days Benzathine penicillin G IM x 1 dose (if adherence is a concern)	Use first line for majority of GAS cases	• Excellent safety profile • May cause mild diarrhea
Cephalexin, clindamycin* or clarithromycin* x 10 days	Patients with type 4 penicillin allergy or amoxicillin hypersensitivity (e.g. rash)	• Clindamycin: C. difficile infection • Clarithromycin: QT prolongation risk • * Clarithromycin or clindamycin (or azithromycin) can be used in type 1 anaphylactic reaction to penicillin
Amoxicillin/clavulanic acid x 3 days Clindamycin x 3 days	Refractory infections to first-line agents	• Take with meal to reduce GI symptoms

MONITORING PARAMETERS

Parameters	Degree of Change	Timeframe
Symptoms	Improvement	Monitor for worsening symptoms Bacterial Pharyngitis: • Fever <48 h • Pain and fatigue 48-72 h • Symptomatic resolution is usually seen in 4-5 days, but 10-day therapy is used to prevent ARF Viral Bronchitis: • Improvement over 5-7 days • Resolution of cough within 10-14 days, but up to 21-28 days or more in some cases
Drug side effects	Minimize	Throughout therapy prevent and manage adverse drug reactions from antibiotics

TAKEAWAYS

- URTIs are the most common cause of acute illness in adults and children, predominantly caused by viruses
  - Symptom relief is the cornerstone of therapy
- The Modified Centor Score is used to calculate the risk of streptococcal pharyngitis and help guide whether diagnostic testing and/or antibiotic treatment is appropriate

- Bacterial pharyngitis treatment:
  - Children: Start within 9 days. Abx duration should be 10 days to eradicate GAS
    - Most patients: penicillin V, amoxicillin 50mg/kg daily, penicillin G benzathine (if adherence concerns)
    - Non-anaphylactic hypersensitivity reactions to penicillin: amoxicillin challenge or cephalexin 20mg/kg/ dose twice daily
    - Documented type 1 hypersensitivity or clear history of anaphylaxis to penicillin: azithromycin, clarithromycin or clindamycin
  - Adults: Amoxicillin or penicillin V is the mainstay of therapy to eradicate GAS
    - Type 4 hypersensitivity to amoxicillin: cephalexin, clindamycin or clarithromycin for 10 days
    - Refractory infections to first-line agents: amoxicillin/clavulanic acid or clindamycin for 3 days

CASE 1

TK, an 8-year-old boy, is complaining of a sore throat in ER. TK's mom reports that TK has had an intermittent fever of 39°C over the past day which was being treated with acetaminophen suspension; he is unable to swallow solid dosage forms. Other symptoms TK is experiencing include lethargy and decreased appetite. When asked, TK denies any cough or shortness of breath. TK has a history of otitis media from a few months ago. His vaccinations are up-to-date. He is allergic to penicillin (GI upset). He lives at home with his parents and is attending a local school. Today, his BP is 108/75 mmHg, his pulse is 90 bpm, his RR 20 bpm, his temperature is 38.8°C, and he weighs 31 kg.

1. What would confirm a diagnosis of GABHS acute pharyngitis?
   a) Fever
   b) Modified Centor Score >3
   c) Positive throat culture
   d) Chills and sore throat

CASE 1

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2. What is the most appropriate antibiotic therapy for TK?
    a) Azithromycin
    b) Amoxicillin
    c) Cephalexin
    d) Clarithromycin

CASE 1

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3. How should TK be monitored to determine if the infection is resolved?
    a) Modified Centor Score = 0
    b) Symptom resolution
    c) Repeat throat culture
    d) Absence of skin rash

CASE 2

DJ is a 22-year-old female pharmacy student at the University of Toronto who is complaining of a productive cough, mild headache and sore throat on waking for 5 days. DJ came into her student health centre because she was concerned her symptoms were persistent and that her partner got her sick. She denies any fever, chills or shortness of breath. DJ has had mild acne for 8 years for which she uses benzoyl peroxide 2.5% topically daily PRN. She has no allergies to medications. She is a social smoker. Today, upon examination, she is afebrile and has no cervical lymphadenopathy.

1. What of the following is the most reasonable diagnostic test to consider for DJ?
   a) Chlamydia and gonorrhea NAAT
   b) Chest x-ray
   c) Group A streptococcal rapid antigen test
   d) Vital signs and chest auscultation

CASE 2

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2. Assuming this test is negative, what are is best therapeutic recommendation for DJ right now?
   a) Clarithromycin 500mg PO BID x 7 days
   b) Echinacea
   c) Acetaminophen 1000mg PO Q6H PRN
   d) Fluticasone 125 mcg 2 puffs BID via valved holding chamber

CASE 2 - QUESTIONS

DJ is a 22-year-old female pharmacy student at the University of Toronto who is complaining of a productive cough, mild headache and sore throat on waking for 5 days. DJ came into her student health centre because she was concerned her symptoms were persistent and that her partner got her sick. She denies any fever, chills or shortness of breath. DJ has had mild acne for 8 years for which she uses benzoyl peroxide 2.5% topically daily PRN. She has no allergies to medications. She is a social smoker. Today, upon examination, she is afebrile and has no cervical lymphadenopathy.

3. Which of the following counselling points is important to convey to DJ prior to leaving?
   a) Cough can persist up to 3-4 days after an episode of acute bronchitis
   b) Pneumonia is a possible complication of bronchitis
   c) She should consider honey and warm beverages to reduce her cough frequency
   d) Codeine OTC is a safe and effective remedy for chronic cough

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CHANGE LOG

July 2025

- Formatting edits made throughout; legend and references updated
- Slide 4: Edited definition of acute bronchitis
- Slide 10: Tables adjusted to better align with the Canadian guidelines
- Slide 13: Added note on lozenge/lidocaine use
- Slide 16: Added note on corticosteroid use in bacterial pharyngitis
- Slide 17: Added drug agents
- Slide 19: Modified the Takeaway slide
- Slide 21: Modified option D

July 2025

- Slides 17 and 19: Removed mention of cefpodoxime as available on the Canadian market for veterinary use only

October 2025

- Formatting changes throughout, reorganized content for better flow
- Slide 4: Added new definitions
- Slide 6: Combined two slides into one
- Slide 7: Added disease states to differential diagnosis
- Slide 8: Removed chronic bronchitis; clarified viral vs. GAS pharyngitis and children's acute bronchitis symptoms
- Slide 10: Added pharyngitis etiology
- Slide 14: Changed age to 3 years
- Slide 16: New child-focused slide
- Slide 17: Included clindamycin × 3 days for refractory infections

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 17: Elaborated on treatment options (added Pen V to first line options and included alternatives for anaphylactic penicillin allergy)
- Slide 19: Updated takeaway slide
- Slide 26: Updated references